

Does a
Momentum-based
Stock Selection
Strategy Outperform
the S&P 500?

Agenda

Define Strategy

Why This Matters to You

Methods

Individual Stock Analysis

Strategy vs. Benchmark Analysis

Recommendation

What is a Momentum-based Stock Selection Strategy?

Momentum Strategy: Buying the top-performing stocks over a short period of time, and selling those that are underperforming.

Why This Matters to You

- Investing in index funds is the default recommendation
- Can a momentum-based strategy deliver better risk-adjusted returns than just holding an index fund?

Methods

- Data selected: 15 large-cap stocks across sectors (tech, financials, healthcare, energy, etc...)
- Timeline: 2022 - 2026
- Process: Pulled Yahoo Finance data → stored it in SQL database → queries out data to be analyzed → analyzed data in Python
- Strategy Rule: Rank the 15 stocks' performances every 6-months, select the top 5 stocks

Backend Code

Risk Metrics: Sharpe Ratio, Max Drawdown

```
import numpy as np

sharpe_ratios = (daily_returns.mean() / daily_returns.std()) * np.sqrt(252)
sharpe_ratios.sort_values(ascending = False)
```

Python

```
# Running max: highest value reached so far at each point in time
running_max = cumulative_returns.cummax()

# Drawdown: how far below that peak you are at each point in time
drawdown = (cumulative_returns - running_max) / running_max
```

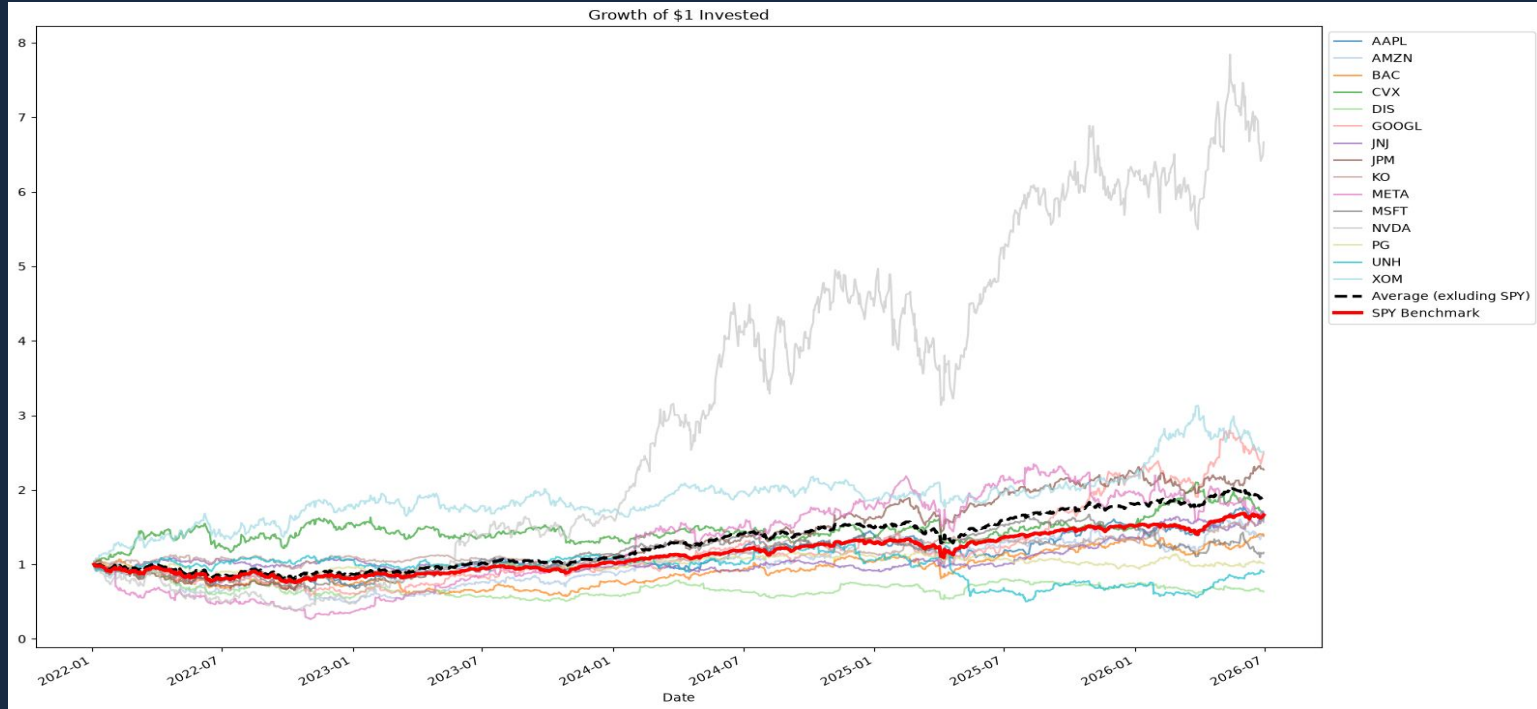
```
# Max drawdown
max_drawdown = drawdown.min()
max_drawdown.sort_values(ascending = False)
```

Python

```
# Create horizontal summary table
summary_table = pd.DataFrame({
    'Total Return': cumulative_returns.iloc[-1] - 1,
    'Sharpe Ratio': sharpe_ratios,
    'Max Drawdown': max_drawdown
})
```

- Building analysis

Individual Stock Analysis



Definitions

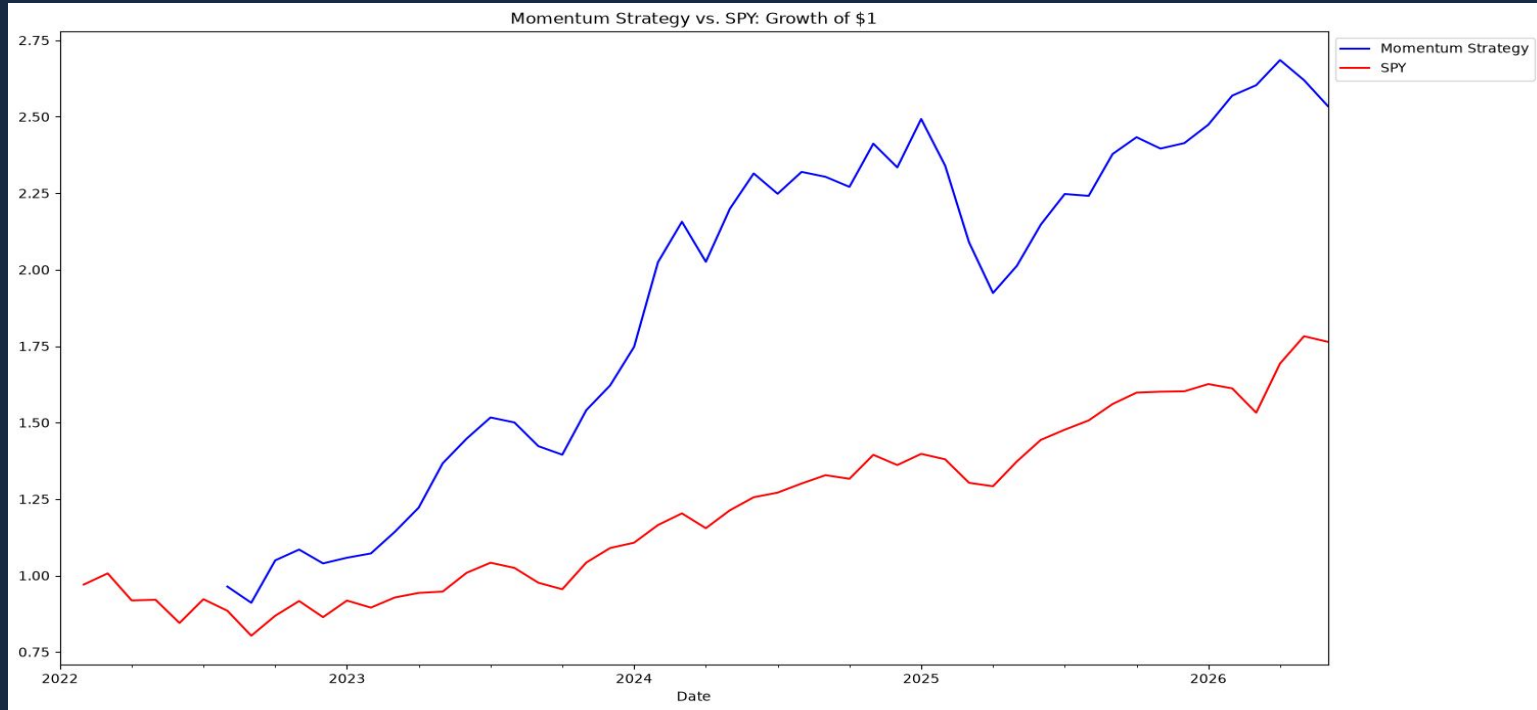
- Sharpe Ratio: How good is an investments return relative to its risk
- Max Drawdown: What is the worst percentage drop I could have faced holding this stock
- A risk-adjusted analysis is a better indicator of performance



Individual Stock Analysis [Risk-adjusted]

	Total Return	Sharpe Ratio	Max Drawdown
AAPL	62.60%	0.53	-33.40%
AMZN	39.90%	0.39	-51.70%
BAC	38.10%	0.4	-46.60%
CVX	66.40%	0.58	-24.90%
DIS	-36.70%	-0.19	-49.80%
GOOGL	148.70%	0.79	-43.60%
JNJ	68.30%	0.75	-18.40%
JPM	127.00%	0.86	-37.90%
KO	56.50%	0.69	-17.30%
META	67.80%	0.48	-73.60%
MSFT	15.70%	0.26	-34.50%
NVDA	566.40%	1.07	-61.60%
PG	1.00%	0.1	-23.80%
SPY	66.10%	0.73	-24.50%
UNH	-10.30%	0.1	-61.40%
XOM	151.40%	0.91	-20.50%

Momentum Strategy vs. Benchmark



Momentum Strategy vs. Benchmark [Risk-adjusted]

	Strategy	SPY
Total Return	153.50%	76.40%
Sharpe Ratio	1.29	0.89
Max Drawdown	-22.80%	-20.20%

- The Momentum Strategy beat the SPY on returns and Sharpe, while having similar drawdowns

Caveats

- Small stock pool of 15 stocks
- The stocks selected were already big winners in 2026
- Short timeline of 2022-2026
- Transaction costs and taxes are not factored into the results



Ticker	Number of Times Selected
NVDA	31
JPM	28
META	28
GOOGL	22
AMZN	15
MSFT	15
XOM	15
BAC	14
CVX	14
JNJ	12
AAPL	11
KO	11
UNH	10
DIS	6
PG	3

Recommendation

- This momentum approach does outperform the SPY, but given the previous risks that I mentioned before, I would not recommend using this strategy.



What I Learned

1.

The
process of
doing an
analysis

2.

Explaining
a technical
project

3.

Presenting
flaws in a
project

4.

Using AI as
a tool, not
a solution

Thank You